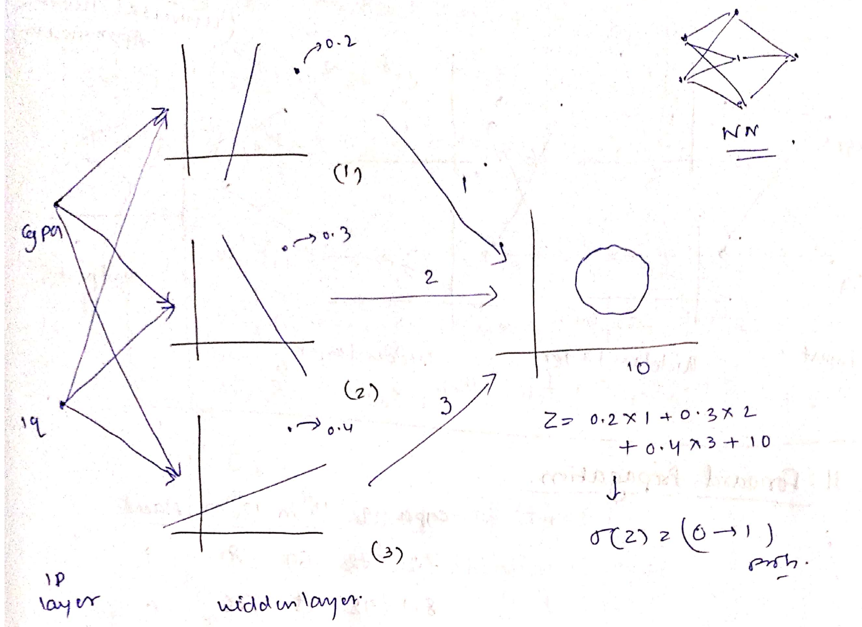
Adding nodes in hidden

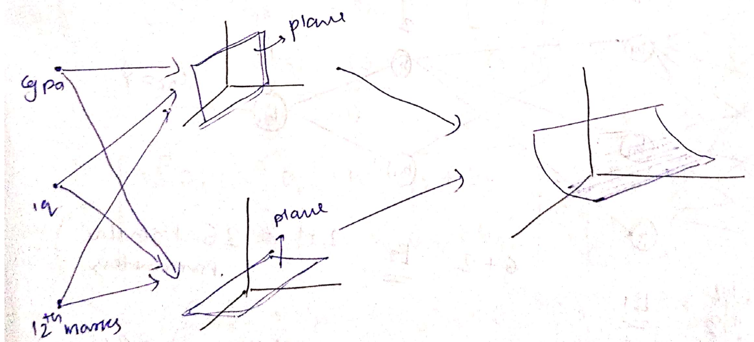


$Z = 0.2\times1 + 0.3\times2 + 0.4\times3 + 10$

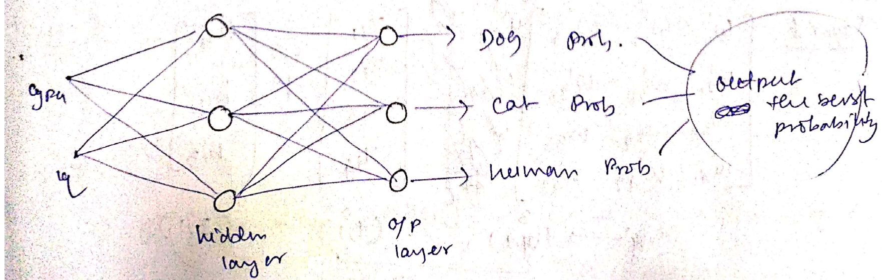
$\downarrow$

$\sigma(z) = (0 \rightarrow 1)$ prob.

Adding Nodes in Input.

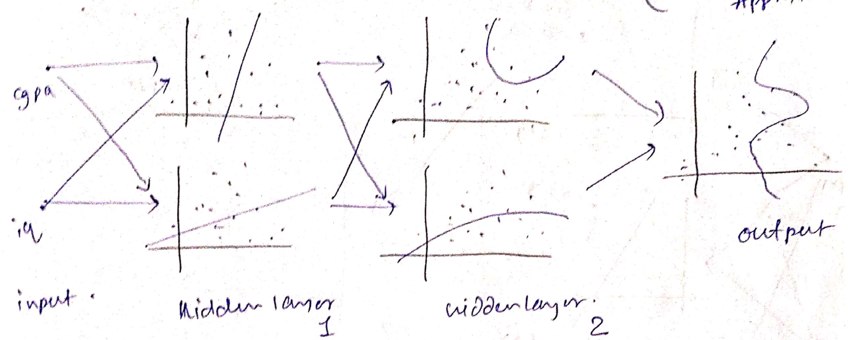


Adding nodes in output ~~layer~~ node.



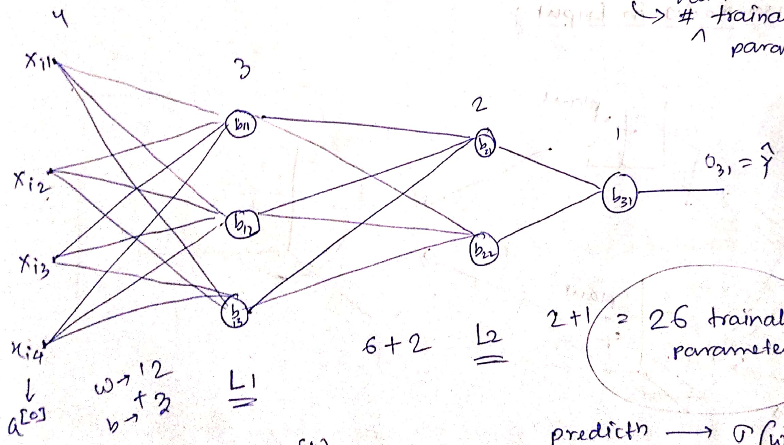
Deep Neural Network (Adding layers)

$\rightarrow$ UFA (universal function Approximator)



Forward Propagation.

cgpa	iq	10th m	12th m	placed
7.2	72	69	81	1
8.1	92	75	76	0



$w \rightarrow 12$

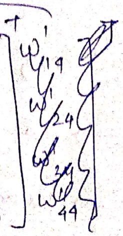
$b \rightarrow 3$

L1   6 + 2   L2   2 + 1 = 26 trainable parameters

$\rightarrow$ # no. of trainable parameters

predict $\rightarrow \sigma(w^Tx + b)$

layer 1



$$\overbrace{\begin{bmatrix} W^1_{11} & W^1_{12} & W^1_{13} \\ W^1_{21} & W^1_{22} & W^1_{23} \\ W^1_{31} & W^1_{32} & W^1_{33} \\ W^1_{41} & W^1_{42} & W^1_{43} \end{bmatrix}^T}^{W^{(1)}} \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$(4,3) \xrightarrow{T} (3,4) \quad (4,1) \quad (3,1)$

$(3,1)$

$$= \begin{bmatrix} w'_{11}x_{i1} + w'_{21}x_{i2} + w'_{31}x_{i3} + w'_{41}x_{i4} \\ w'_{12}x_{i1} + w'_{22}x_{i2} + w'_{32}x_{i3} + w'_{42}x_{i4} \\ w'_{13}x_{i1} + w'_{23}x_{i2} + w'_{33}x_{i3} + w'_{43}x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$\Rightarrow \sigma\left(\begin{bmatrix} w'_{11}x_{i1} + w'_{21}x_{i2} + w'_{31}x_{i3} + w'_{41}x_{i4} + b_{11} \\ w'_{12}x_{i1} + w'_{22}x_{i2} + w'_{32}x_{i3} + w'_{42}x_{i4} + b_{12} \\ w'_{13}x_{i1} + w'_{23}x_{i2} + w'_{33}x_{i3} + w'_{43}x_{i4} + b_{13} \end{bmatrix} \right)$$

$$= \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} \rightarrow \text{output of layer 1.}$$

$= a^{[1]} \rightarrow$ activation of layer 1

layer 2.

$$\overbrace{\begin{bmatrix} w^2_{11} & w^2_{12} \\ w^2_{21} & w^2_{22} \\ w^2_{31} & w^2_{32} \end{bmatrix}^T}^{W^{[2]}} \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$(3,2) \xrightarrow{T} \underbrace{(2,3) \quad (3,1)}_{(2,1)} \quad (2,1)$

$$= \begin{bmatrix} w^2_{11}O_{11} + w^2_{21}O_{12} + w^2_{31}O_{13} + b_{21} \\ w^2_{12}O_{11} + w^2_{22}O_{12} + w^2_{32}O_{13} + b_{22} \end{bmatrix} \cancel{+ \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}}$$

$$\Rightarrow \sigma\left(\begin{bmatrix} w^2_{11}O_{11} + w^2_{21}O_{12} + w^2_{31}O_{13} + b_{21} \\ w^2_{12}O_{11} + w^2_{22}O_{12} + w^2_{32}O_{13} + b_{22} \end{bmatrix} \right) = \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} \rightarrow \text{Output of layer 2}$$

$= a^{[2]} \rightarrow$ activation of layer 2

layer 3

$$\overbrace{\begin{bmatrix} w^3_{11} \\ w^3_{21} \end{bmatrix}^T}^{W^{[3]}} \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$(2,1) \xrightarrow{T} \underbrace{(1,2) \quad (2,1)}_{} \quad (1,1)$

$$\Rightarrow \sigma\left(\begin{bmatrix} w^3_{11}O_{21} + w^2_{21}O_{22} + b_{31} \end{bmatrix} \right) = O_{31} = \hat{y}_i \rightarrow \text{final output.}$$

$= a^{[3]} \rightarrow$ activation 3.

~~$a^{[0]} W^{[1]}$~~

$$a^{[1]} = \sigma(a^{[0]}W^{[1]} + b^{[1]})$$
$$a^{[2]} = \sigma(a^{[1]}W^{[2]} + b^{[2]})$$
$$a^{[3]} = \sigma(a^{[2]}W^{[3]} + b^{[3]})$$

final o/p

$$\sigma\Bigg(\underbrace{\sigma\Big(\underbrace{\sigma(a^{[0]}W^{[1]} + b^{[1]})}_{a^{[1]}} W^{[2]} + b^{[2]}\Big)}_{a^{[2]}} W^{[3]} + b^{[3]}\Bigg)$$

$\uparrow a^{[3]}$

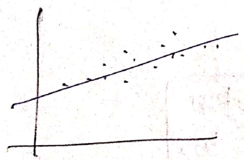
18/07/25

What is Loss fn?

Loss fn is a method of evaluating how well your algorithm is modelling your dataset.

high → poor
small → great

$L(\text{parameters}) \leftarrow f(x) = 3x^2 + 2$



G.D.

$$L = (y_i - \hat{y}_i)^2$$

$\hat{y}_i = mx_i + b$

$L(m,b) = \min.$

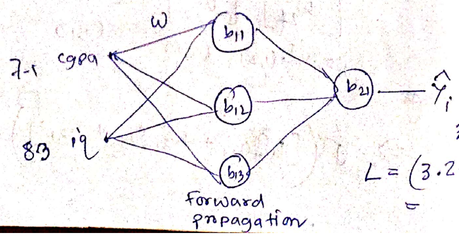
Why is Loss fn important?

[You can't improve what you can't measure]
Peter Drucker

Loss function → eye of ML algorithm.

Loss fn in Deep Learning?

cgpa	iq	package (LPA)
7.1	83	3.2
8.5	91	4.5
6.3	102	6.1
5.1	89	2.7



$L = (3.2 - 3.$

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